FOCAL

Vision testing you can do anywhere, and an organization can trust.

Business Model Presentation · MENG 570 · Team 5

Haydn Stucker
CEO

Burak Donbekci
CTO

Hadley McCormack
CFO

Fan Zhang
CCO

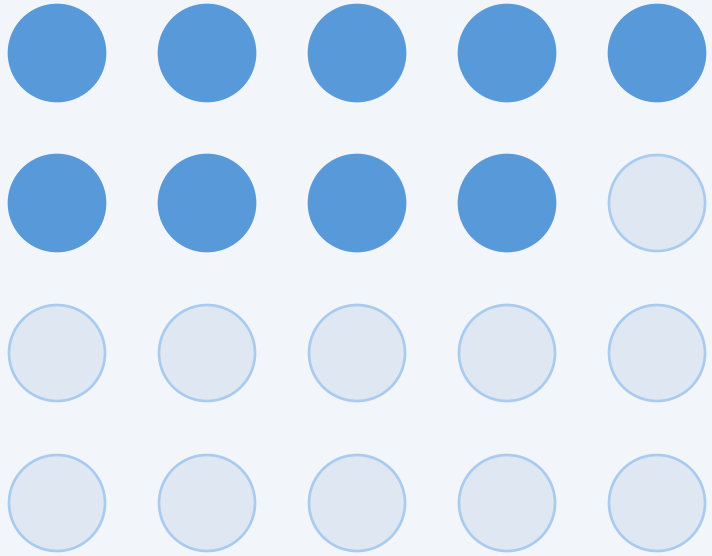
Qingyi Hu
CPO

Vision loss is common and often unaddressed

2.2 billion

people live with a vision impairment

At least 1 billion of those cases were preventable or not yet addressed.



~45% preventable or unaddressed

Early detection needs routine checks. Routine checks are hard to get.

Source: WHO, World report on vision (2019) and Blindness and vision impairment fact sheet.

Convenient or thorough: pick one

WHO FEELS IT

Individuals
adults whose exam has lapsed

Parents
relying on school screenings

Organizations
licensing agencies, rental companies, insurers

01

Access is hard and outdated

Even a basic check means a clinic visit and a wait.

02

Changes Are Hard to Track

One snapshot can't show change. Home charts get memorized.

03

Online tests are hard to trust

Relationships with healthcare providers and organizations take time to build.

What we heard in 10 interviews

10

customer discovery interviews

4 personas · 5 virtual, 5 in person

Avoidance is not only logistical

"Maybe I'm fine" is more comfortable than knowing.

Urgency comes from outside

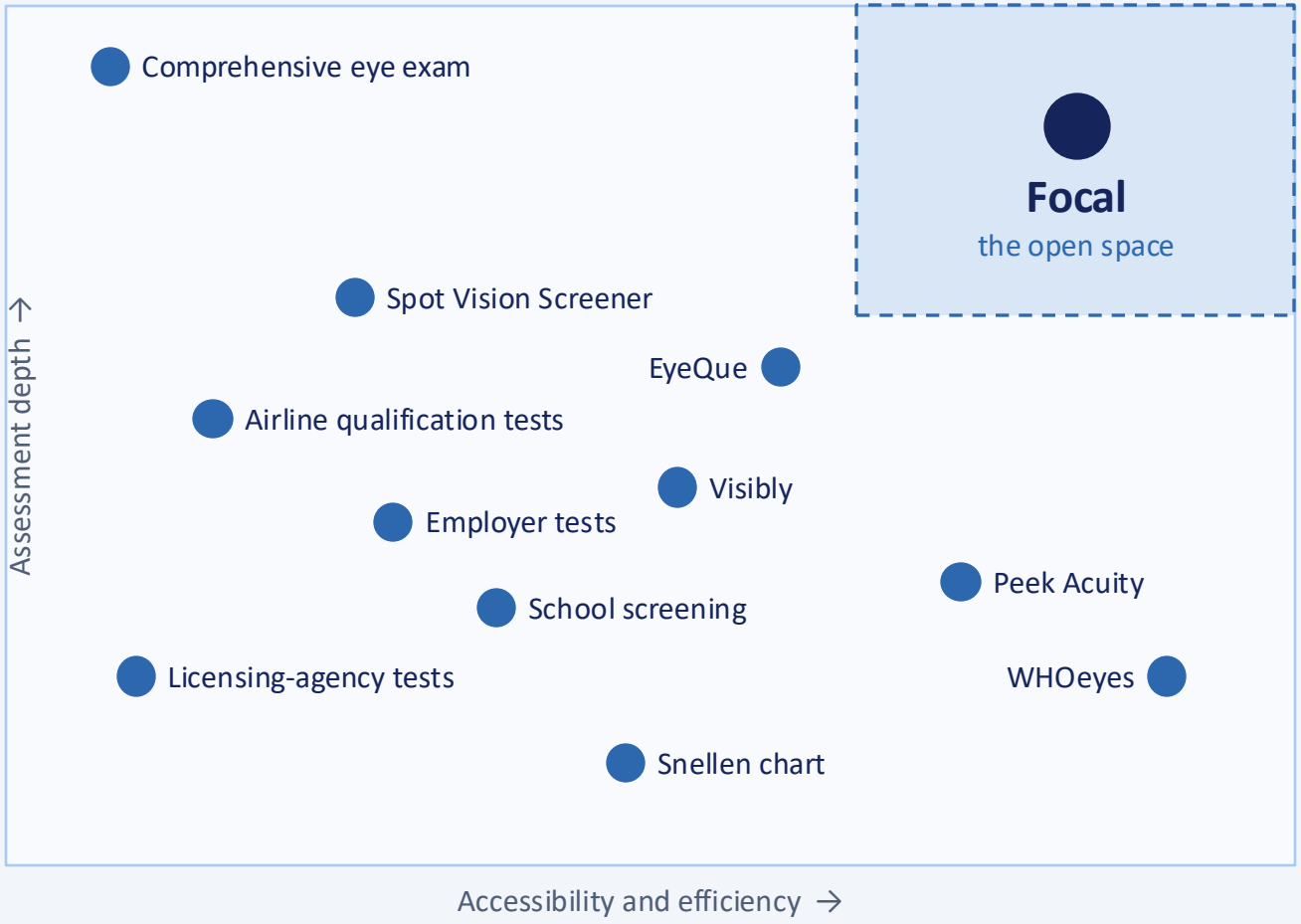
An expiring prescription or a school letter, not symptoms.

Home charts get memorized

Children learn the letters, so repeat tests tell you nothing.

Small, convenience sample. Our licensing supervisor persona was reached only through AI-simulated interviews.

11 alternatives, none in the top right



WHERE EACH FALLS SHORT

Visibly
Ages 22 to 40 only; acuity only

EyeQue
Needs $129 phone hardware

Peek Acuity, WHOeyes
Acuity only

Clinic exam
Needs an appointment and staff

Nothing is both easy and deep.

Early adopters and why they switch

PRIMARY EARLY ADOPTER · RILEY

Adults on screens all day

SWITCHING TRIGGER

Prescription is about to expire

POTENTIAL FUTURE SEGMENT · PAT

Parents of school-age children

SWITCHING TRIGGER

A school screening letter arrives

POTENTIAL FUTURE SEGMENT· ALEX

First-time license applicants

SWITCHING TRIGGER

A DMV vision screening is coming up

THEN WE EXPAND

Individuals and parents
interviewed in person

Pediatric offices and insurers
a young adult leaves a parent's plan

Licensing and rental agencies
validate with real agency staff first

UNIQUE VALUE PROPOSITION

Routine vision testing you can do anywhere, and trust enough to act on.

Accessible

An app, no clinic visit

Adaptive

Varies each test, fewer false positives

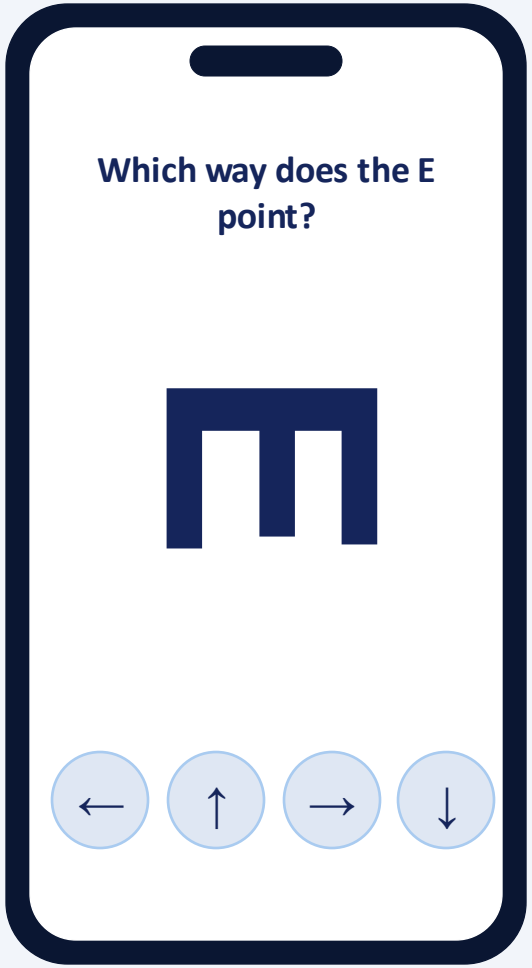
Actionable

Tells you if you need an exam

The solution: test, track, guide

1. **Test**
   An adaptive test that changes every time
2. **Track**
   A baseline, then your trend
3. **Guide**
   A calm answer: do you need an exam?
4. **Share**
   Turn a recommendation into professional follow-up

FIRST MVP (proposed) Low-fidelity result screens, tested with real users



Illustrative mockup

Where Focal is different

	Visibly	EyeQue	Peek Acuity	WHOeyes	Clinic exam	Focal (goal)
No extra hardware needed						
Goes beyond acuity						
Tracks change over time						
Accepted by organizations						
Regulatory readiness						

Only Focal aims at depth, tracking and trust together.

Competitor ratings from our Harvey Ball analysis. Focal column = goals, not measured results.

Unfair Advantage

01

Adaptive-testing method

AI personalizes each test. A patent candidate.

02

Longitudinal data

Every repeat test makes our predictions better.

03

Compliance pipeline

Validated workflows and partnerships that could create switching costs over time.

WHAT WE DON'T CLAIM

~~First-mover advantage~~

~~Passion~~

~~Features~~

How we will know it works

$1M

ARR within 3 years

Our minimum success criteria (MSC)

FOUR LEADING INDICATORS

- **Results users trust**
  Trust is the first thing we test
- **Users who act on the advice**
  Do they book an exam when told to?
- **Repeat-test rate**
  Tracking has real value
- **Accuracy vs. a clinical chart**
  Fewer false positives is the UVP

Channels, costs and revenue

CHANNELS (DIRECTIONAL)

- **Free "Will I pass my DMV eye test?" checker**
- **School nurses and pediatric offices**
- **Insurer member apps**
- **Rental app check-in**

COSTS

- AI development and validation
- Regulatory and compliance
- Secure health-data hosting

REVENUE

- Free baseline test for individuals
- Premium personal tracking
- Organization licensing or per-test fees

Key takeaways

1 A real problem

Vision testing is convenient or thorough, never both.

2 A clear first customer

Individuals and parents with a trigger: an expiring prescription, a school letter.

3 A differentiated answer

Test, track and guide: the open space in the market.

Focal: Lean Canvas

Focal Technologies · Mission: creating simple and interpretable technology for wellness · Haydn Stucker CEO · Burak Donbekci CTO · Hadley McCormack CFO · Fan CCO · Qingyi CPO

PROBLEM

- Access to vision testing is difficult and outdated
- Existing tests are poor and not predictive
- Telehealth is not built for routine checks

EXISTING ALTERNATIVES

- Visibly, EyeQue, Peek Acuity, WHOeyes
- Clinic exam, licensing-agency tests

SOLUTION

- App to test vision, adaptive and AI-enhanced
- Baseline and trend tracking
- Next-step guidance; share with an optometrist

KEY METRICS

- MSC: $1M ARR in 3 years
- Results users trust
- Users who act on advice
- Repeat-test rate; accuracy vs. clinical chart

UNIQUE VALUE PROPOSITION

Routine vision testing you can do anywhere, and trust enough to act on.

Accessible · Adaptive · Actionable

UNFAIR ADVANTAGE

- AI-personalized testing (patent candidate)
- Longitudinal vision dataset
- Compliance-check pipeline for agencies

CHANNELS

- Free DMV eye-test checker
- School nurses, pediatric offices
- Insurer member apps
- Rental app check-in

CUSTOMER SEGMENTS

- Individual users and parents
- Pediatric development
- Insurance providers
- Vehicle renting and licensing agencies
- Healthcare and lens providers

EARLY ADOPTERS

- Riley: adults with an expiring prescription
- Pat: parents with a school letter
- Alex: first-time license applicants

COST STRUCTURE

- AI model development and clinical validation
- Regulatory and compliance work
- Secure hosting for health data

REVENUE STREAMS

- Free baseline test for individuals
- Premium personal tracking
- Organization licensing or per-test fees (pricing TBD)

Backup: risks we are watching

RISK	OUR DRAFT RESPONSE
Optometry pushback	Position as screening that refers people to an exam
Licensing demand is unproven	Only virtual interviews so far. Validate with a real agency employee
Regulators distrust remote results	Validate against clinical charts before claiming accuracy
Health-data privacy and security	Collect the minimum, host securely, build in compliance
Big eye-care companies copy us	Move fast on our dataset and organization acceptance